Title: Walking Surface Analysis Using StappOne Pressure Insoles

Introduction:

This project explores the application of wearable pressure sensor data in urban walking research. Using data collected from StappOne shoe insole sensors, I compared gait patterns across two common urban surfaces: grass and concrete. Using Python (pandas and matplotlib), I analyzed total pressure over time, detected footsteps through peak detection, and examined average pressure distribution across 12 sensor zones. A key finding was that concrete produced notably denser and more frequent pressure peaks than grass, suggesting meaningful differences in how the human body interacts with different walking surfaces.

Methods

The CSV data was first filtered to remove corrupt rows, then separated by sole_id into left foot (sole_id 2) and right foot (sole_id 1) for each surface. Timestamps were converted from Unix milliseconds to seconds starting from zero. Total pressure per timestep was calculated by summing all 12 sensor values. Step detection was implemented using a custom peak detection algorithm — a point was classified as a step if its pressure value exceeded both its immediate neighbors and a minimum threshold of 4000 units, filtering out sensor noise. Cadence was then calculated as steps per minute using total recording duration. Finally, average pressure per sensor zone was computed across each surface to compare spatial pressure distribution.

Figure 1 shows total pressure over time for both feet on grass. The peaks are rounded and smooth, reflecting the soft, uneven nature of the surface. A clear asymmetry is visible between left and right foot, with the left foot consistently carrying higher pressure.

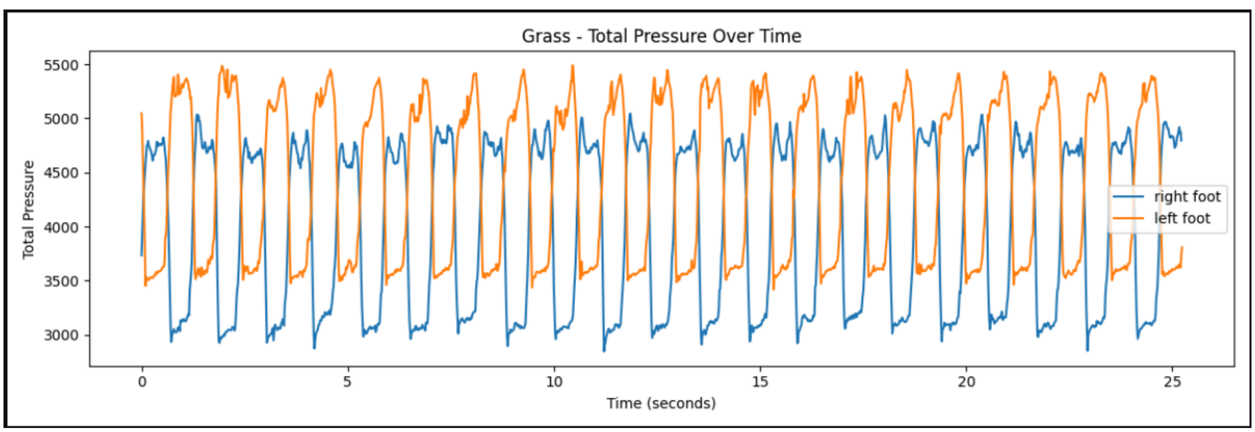


Figure 2 shows the same for concrete. The peaks are noticeably sharper and more densely packed, indicating faster, more confident steps on the rigid surface. Left and right foot pressure is more balanced compared to grass.

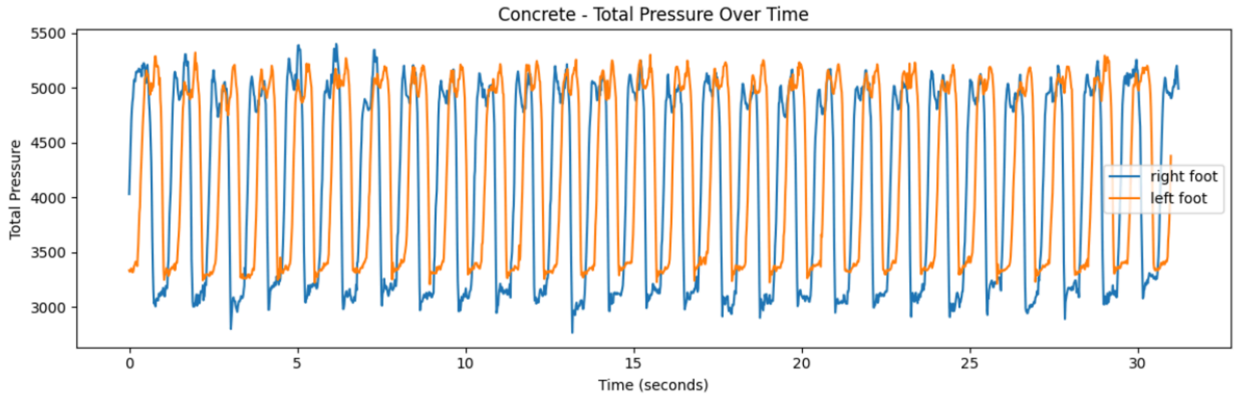


Figure 3 compares average pressure across all 12 sensor zones for both surfaces. Sensors 2, 4, 6 and 7 consistently show the highest pressure on both surfaces, corresponding to the midfoot and forefoot regions. Overall pressure values are slightly higher on grass across most sensors.

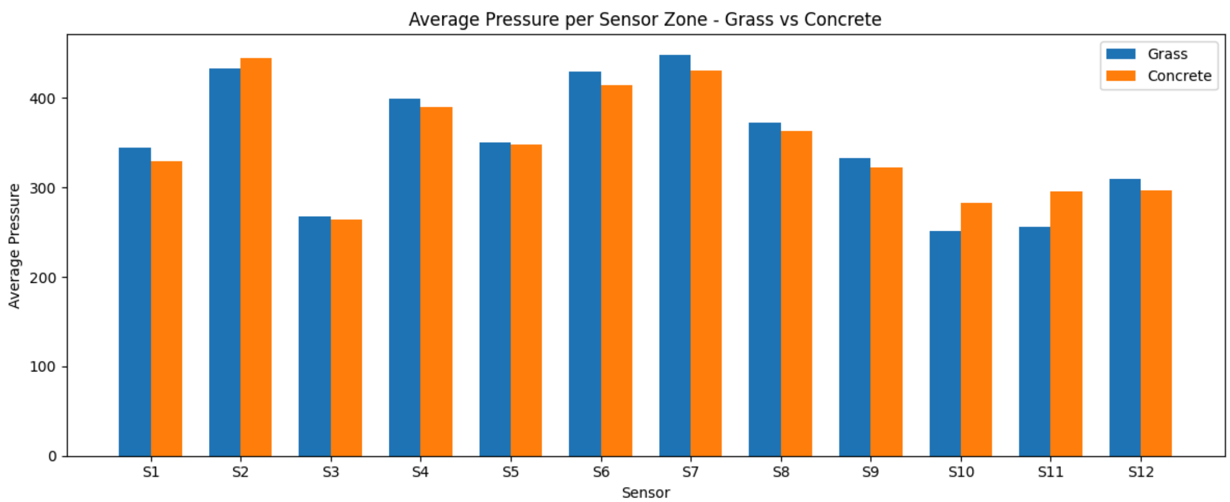
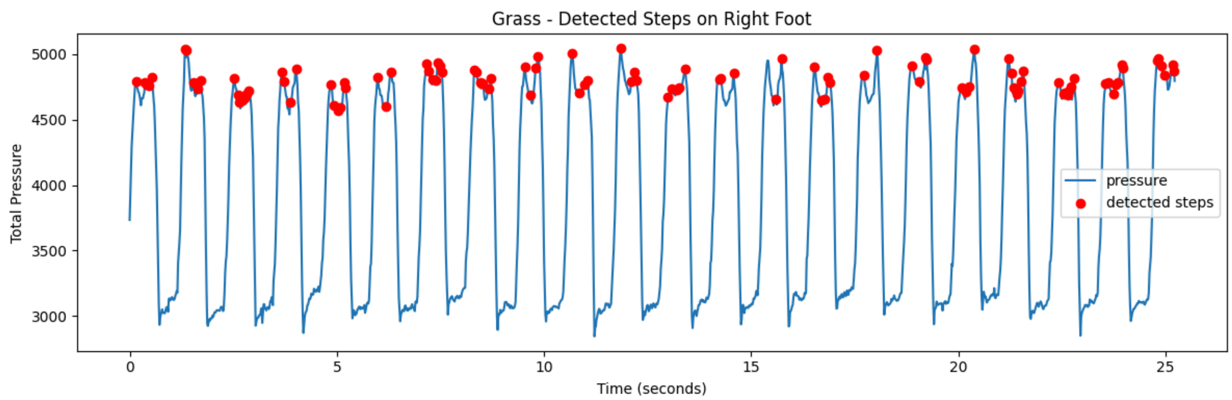


Figure 4 shows detected steps marked on the grass pressure signal. The algorithm successfully identifies most steps, though some peaks contain multiple detections due to small sub-peaks within a single step — a limitation of the simple threshold-based approach.



Discussion

The differences observed between grass and concrete reflect the distinct biomechanical demands of each surface. Grass, being soft and uneven, causes slower walking with rounder pressure peaks and greater left/right asymmetry — the foot constantly adjusts to the irregular terrain, engaging stabilizing muscles more actively. Concrete, being flat and rigid, enables faster and more efficient walking with sharper, more symmetric pressure patterns. However, the hard surface transmits greater impact force to the knees and joints with each step.

Both surfaces offer distinct health implications for urban walkers. Grass encourages slower, more varied movement that engages stabilizing muscles, potentially improving balance and coordination. Concrete enables faster movement but may contribute to joint stress over long term daily exposure. These findings suggest that urban design decisions — such as the ratio of grass paths to concrete sidewalks — may meaningfully affect the health outcomes of city walkers.

ML thoughts

For future work, a machine learning classifier trained on larger datasets from multiple walkers across different surfaces could potentially identify surface type automatically from pressure patterns. The distinctive peak shapes observed in this analysis — rounded for grass, sharp for concrete — could serve as key features for such a model.